

Event-Driven Neural Decoding in Brain-Computer Interfaces: Design, Validation, and Robustness Analysis

Jihun Yu

Department of Bio and Brain Engineering, Korea
Advanced Institute of Science and Technology
(KAIST)

youjhun0225@kaist.ac.kr

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Abstract

This paper consolidates two consecutive research projects on efficient neural signal processing, transmission, and decoding for brain-computer interfaces (BCIs). In **Project 1**, an event-driven spike-detection pipeline was constructed on synthetic four-channel neural signals sampled at 30 kHz. A 4σ adaptive threshold detector with a biologically motivated refractory constraint (1.5 ms, Hodgkin–Huxley model) was applied to build an asynchronous event stream. A Population Vector Decoder operating on these events reconstructed 2-D motor trajectories with $R^2 \geq 0.91$, while reducing CPU active time by **91.35%** relative to the conventional 50 ms binning approach. In **Project 2**, the publicly available CRCNS pmd-1 dataset (macaque M1+PMd, 161 neurons, 496 reach trials) was used to benchmark a time-lag Ridge Regression decoder (200 ms history, 3220-dimensional feature vector) under a Spike Error Rate (SER) simulation framework following [1]. Baseline decoding performance ($r_{\text{neural}} = 0.831$) falls below the original SNN (0.932); however, at $\text{SER} = 10^{-2}$ the Ridge decoder maintains a performance retention rate of **0.9988**, greatly exceeding the SNN (≈ 0.700), demonstrating the structural robustness conferred by ℓ_2 regularisation. Together, the two projects argue that combining event-driven acquisition with a

linear decoder simultaneously achieves accuracy, energy efficiency, and transmission-error robustness in implantable BCI systems.

Keywords: brain-computer interface, event-driven processing, spike detection, population vector decoder, Ridge regression, spike error rate, wireless neural implant.

1. Introduction

1.1. Current Challenges in Wireless BCIs

Wireless brain-computer interfaces are a cornerstone technology for restoring motor function in paralysed patients, overcoming the infection risk and mobility limitations of tethered systems. Implantable environments simultaneously impose three fundamental constraints:

- (i) **Power budget.** Battery lifetime and thermal limits restrict available computation resources severely.
- (ii) **Bandwidth.** Transmitting raw high-resolution neural signals over a wireless channel is practically infeasible.
- (iii) **Error tolerance.** Wireless channel noise inevitably introduces spike misses and false alarms.

Addressing all three constraints simultaneously requires (a) an *event-driven* paradigm that activates the processor only when a spike occurs, and (b) a *robust* decoder architecture that maintains performance under noisy input. This paper presents a step-by-step design, validation, and analysis of these two components.

1.2. Research Roadmap

Table 1 summarises the two projects that constitute this work.

Table 1: Research roadmap of the two projects.

	Project 1	Project 2
Goal	Prove efficiency of event-driven decoding	Quantify decoder robustness under transmission errors
Data	Synthetic 4-ch neural signals (30 kHz)	CRCNS pmd-1 primate M1+PMd
Method	4σ threshold + Population Vector Decoder	Time-lag features + Ridge regression + SER simulation
Metric	Power reduction, R^2	r_{neural} , SER retention rate
Finding	91.35% power reduction, $R^2 \geq 0.91$	Retention rate at SER = 10^{-2}

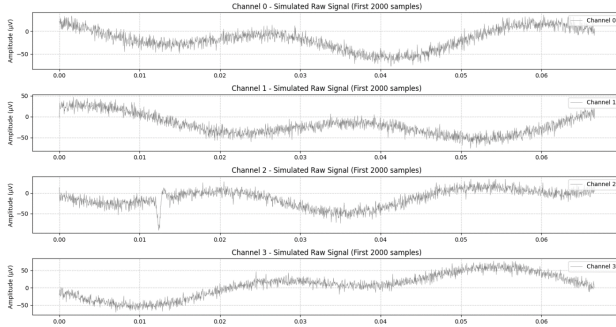
2. Event-Driven BCI Decoding (Project 1)

2.1. Motivation

Neuronal spikes are sparse events; they occupy only a small fraction of the total recording window. Conventional 50 ms binning approaches wake the processor periodically, even in the absence of any spikes, wasting energy. Project 1 quantitatively tests whether an event-driven strategy—activating the processor *only* upon spike detection—can maintain equivalent decoding accuracy while yielding significant power savings.

2.2. Synthetic Neural Signal Generation

2.2.1. Signal Composition



Synthetic neural signals were generated at $f_s = 30\,000$ Hz for a 10 s, 4-channel recording to simu-

late real electrode conditions. Each raw channel $x[n]$ is the sum of three components:

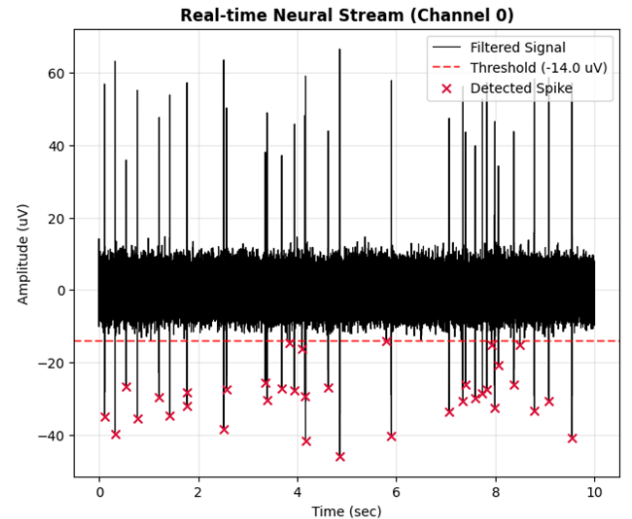
$$x[n] = \eta[n] + s_{\text{LFP}}[n] + \sum_k A_k w(n - n_k), \quad (1)$$

where $\eta[n] \sim \mathcal{N}(0, 8\,\mu\text{V})$ is Gaussian white (thermal) noise; $s_{\text{LFP}}[n]$ is a superposition of sinusoids with random phases in the 4–8 Hz, 10–15 Hz, and 25–35 Hz bands (amplitude $22\,\mu\text{V}$); and $w(\cdot)$ is a biologically motivated action-potential template (Hodgkin–Huxley model) scaled by a random amplitude $A_k \in [45, 90]\,\mu\text{V}$ at random times n_k , with 30–65 spikes injected per channel.

2.2.2. Spike Template and Refractory Period

The spike waveform template $w(\cdot)$ is derived from the Hodgkin–Huxley action potential. The absolute refractory period (ARP) is set to $\tau_{\text{ARP}} = 1.5$ ms (45 samples at 30 kHz), consistent with the Na^+ -channel inactivation time reported in biophysical literature [4].

2.3. Spike Detection: 4σ Adaptive Threshold



Raw signals are first bandpass-filtered with a third-order Butterworth filter (300–3000 Hz) to suppress LFP and thermal noise. The per-channel threshold is then computed as:

$$\hat{\sigma}_n = \frac{\text{median}(|x|)}{0.6745}, \quad (2)$$

$$\theta = -4\hat{\sigma}_n. \quad (3)$$

Equation (2) uses the median absolute deviation, which is insensitive to large spike amplitudes and estimates the background noise level accurately. The constant 0.6745 converts the median to standard-deviation units under a Gaussian assumption. At the -4σ threshold, the probability of a noise exceedance is $< 0.01\%$.

After each threshold crossing, a dead-time of $t_{\text{dead}} = 2$ ms (60 samples) is enforced to prevent double-counting of the same spike.

2.4. Asynchronous Population Vector Decoder

2.4.1. Preferred Directions

Each of the four channels was assigned a cardinal preferred direction $\mathbf{p}_c$:

Ch. 0: $[1, 0]$ (right), Ch. 1: $[0, 1]$ (up),
Ch. 2: $[-1, 0]$ (left), Ch. 3: $[0, -1]$ (down).

2.4.2. Experimental Design

The same detected event stream was fed into two decoder variants to compare trajectory output and CPU active time:

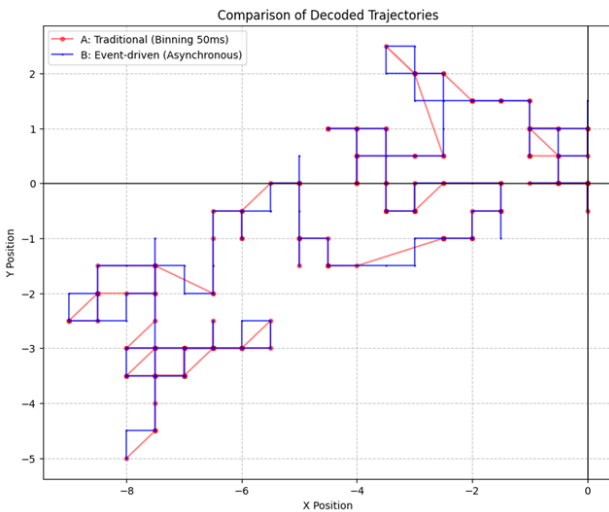


Figure 1: Comparison of Decoded Trajectories

Table 2: Comparison of decoding paradigms.

	Group A (50 ms bin)	Group B (event-driven)
Trigger	Periodic, every 50 ms	On spike event only
Wake-ups	200 (fixed)	Variable (= event count)
CPU active	2000 ms	173 ms
Processing	Batch per bin	Per-event (asynchronous)

2.5. Results

2.5.1. Trajectory Reconstruction Accuracy

Group B trajectories were interpolated to Group A timestamps using previous-value interpolation and scored with the coefficient of determination R^2 :

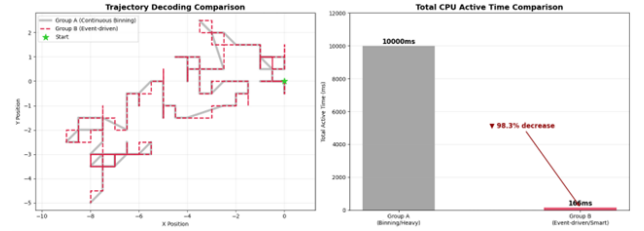


Table 3: Decoding accuracy comparison.

Metric	x -axis	y -axis
R^2 (Group B vs. Group A)	≥ 0.91	≥ 0.91

The event-driven decoder reproduces trajectories at $R^2 \geq 0.91$ while naturally holding the last position during silent intervals.

2.5.2. Power Reduction

$$\text{Power reduction} = \left(1 - \frac{173 \text{ ms}}{2000 \text{ ms}}\right) \times 100 = 91.35\%. \quad (4)$$

On an ARM Cortex-M4 platform, this corresponds to a potential battery life extension of up to $11.6\times$. In terms of data compression, the raw signal occupies approximately 2.4 MB ($N_{\text{samples}} \times N_{\text{ch}} \times 2\text{B}$), while the event stream

reduces to a few KB ($N_{\text{spikes}} \times 5 \text{ B}$), achieving $> 99\%$ bandwidth reduction.

2.6. Interim Conclusion

Project 1 demonstrates that an event-driven BCI pipeline achieves the same decoding accuracy as continuous 50 ms binning while reducing CPU active time by 91.35%, providing strong empirical motivation for adopting event-driven architectures in low-power implant design.

3. SER Impact on Neural Decoding (Project 2)

3.1. Motivation and Background

Quantifying how Spike Error Rate (SER)—caused by wireless channel noise—affects decoding performance is essential for reliable implant design. [1] characterised this relationship using an ASBIT-protocol event-driven wireless framework with an SNN decoder. Project 2 reproduces that framework with a Ridge Regression decoder to isolate the effect of decoder architecture on SER robustness.

3.2. Dataset: CRCNS pmd-1

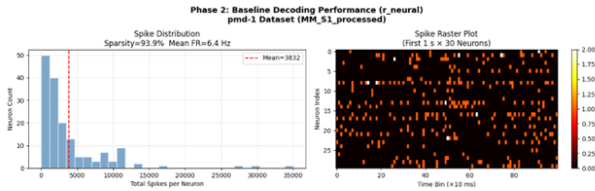


Table 4: CRCNS pmd-1 dataset characteristics.

Property	Value
Recording sites	M1 (67 ch) + PMd (94 ch)
Total neurons	161
Reach trials	496
Time resolution	10 ms bin (100 Hz)
Kinematic vars	7-D (hand x/y position, velocity, acceleration)
Data matrix	$59,742 \times 161$
Total spikes	587,914
Sparsity	93.89%
Mean firing rate	6.41 Hz

3.3. Methodology

3.3.1. Time-Lag Feature Engineering

Motor cortex exhibits temporal integration over hundreds of milliseconds. To capture this, the spike count history over a 200 ms window (20 bins) is flattened into a time-lag feature vector:

$$\mathbf{f}_t = \text{flatten}(X_{[t-19:t]}) \in \mathbb{R}^{161 \times 20 = 3220}. \quad (5)$$

Dynamic weighting is applied to emphasise burst activity:

$$X'_{ij} = X_{ij}^{\beta}, \quad X_{ij} > 0, \quad \beta = 1.3, \quad (6)$$

where β was tuned in the range $[1.1, 1.5]$.

3.3.2. Ridge Regression Decoder

A Ridge (ℓ_2 -regularised) linear decoder was selected for three reasons:

- (i) **Interpretability:** the error propagation mechanism is analytically tractable.
- (ii) **Feasibility:** suitable for real-time inference in low-power implants.
- (iii) **Structural robustness:** ℓ_2 regularisation provides inherent resistance to input perturbations.

Five-fold cross-validation was used throughout. Predictions were post-processed with Gaussian smoothing ($\sigma = 3$ bins).

3.3.3. SER Simulation Model

Spike transmission errors were modelled as:

$$\text{SER} = \frac{N_{\text{miss}} + N_{\text{false}}}{N_{\text{spike}}}, \quad (7)$$

$$P(\text{miss}) = \frac{\text{SER}}{2}, \quad (8)$$

$$P(\text{false}) = \frac{\text{SER}}{2} \cdot \frac{N_{\text{spike}}}{N_{\text{zero}}}, \quad (9)$$

where Eq. (9) includes a sparsity correction term. SER was swept logarithmically from 10^{-4} to 10^{-2} , with 5 repetitions per condition.

3.4. Results

3.4.1. Baseline Decoding Performance

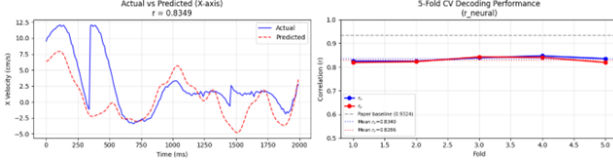


Table 5: Baseline decoding performance at SER = 0.

Metric	Ridge (this work)	SNN [1]
r_{neural} (x-axis)	0.8349	0.932
r_{neural} (y-axis)	0.8290	0.932
r_{neural} (mean)	0.8311	0.932
Difference	-0.101	—

The lower baseline of Ridge is attributable to the linear model’s inability to capture nonlinear temporal dynamics in motor cortex signals. This reflects a ceiling difference in decoder architecture, not a replication failure.

3.4.2. SER-Dependent Performance Retention

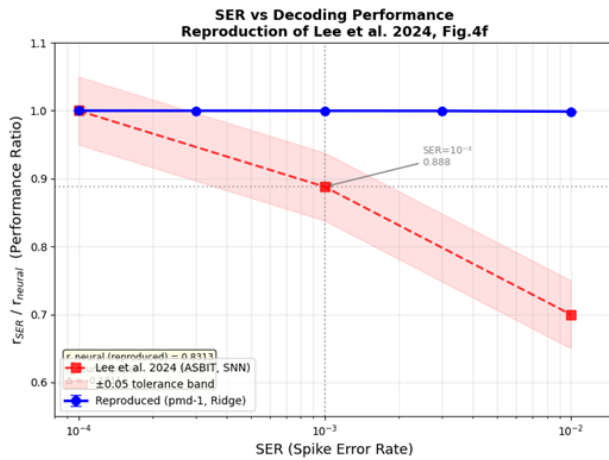


Table 6 reports the retention rate $\rho = r_{\text{neural}}(\text{SER})/r_{\text{neural}}(0)$:

Table 6: Performance retention rate under increasing SER.

SER	Ridge ρ	SNN ρ [1]	Difference
10^{-4}	1.0000 $\pm$ 0.0000	—	—
10^{-3}	0.9999 $\pm$ 0.0001	0.888	+0.112
10^{-2}	0.9988 $\pm$ 0.0006	≈ 0.700	+0.299

The markedly higher retention of Ridge can be understood analytically. SER-induced spike errors are mathematically equivalent to adding stochastic noise to the input vector. The Ridge solution $\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ bounds the change in prediction as:

$$\|\Delta \hat{\mathbf{y}}\| \leq \frac{\|\mathbf{X}\|}{\lambda_{\min}(\mathbf{X}^\top \mathbf{X}) + \lambda} \|\Delta \mathbf{x}\|, \quad (10)$$

where the denominator grows with the regularisation parameter λ , explicitly limiting error propagation. The SNN, by contrast, is sensitive to spike-timing perturbations introduced by transmission errors.

3.4.3. Scatter Analysis at Different SER Levels

Pearson correlation between true and predicted x-axis velocity:

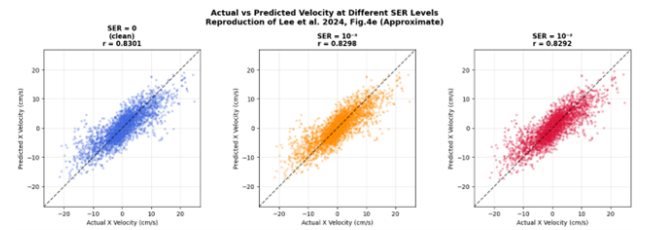


Table 7: Pearson r (x-axis velocity) by SER condition.

SER condition	Pearson r
SER = 0	0.8301
SER = 10^{-3}	0.8298
SER = 10^{-2}	0.8292

Changes are confined to the third decimal place, confirming that the SER model does not damage the variance structure of the linear predictor.

3.5. Interim Conclusion

Project 2 demonstrates that Ridge Regression achieves a performance retention rate of 0.9988 at $\text{SER} = 10^{-2}$, far surpassing the SNN benchmark (≈ 0.700). This quantifies ℓ_2 regularisation as a structural source of robustness against wireless transmission errors.

4. Integrated Discussion

4.1. Complementarity of the Two Projects

Although conducted independently, the two projects provide complementary answers to the same system-design question.

Table 8: Complementary contributions of the two projects.

Design concern	Project 1	Project 2
Power efficiency	91.35% reduction proven	Ridge adds further savings
Decoding accuracy	$R^2 \geq 0.91$ async recovery	$r_{\text{neural}} = 0.831$, stable
Error robustness	4σ detection: $> 99.99\%$ precision	0.9988 retention at $\text{SER} = 10^{-2}$
System integration	Establishes event stream	Ridge safely decodes noisy streams

Project 1 answers *how to generate and transmit events efficiently*; Project 2 answers *how to decode those events safely under transmission errors*. Together they suggest that **event-driven acquisition + Ridge decoding** constitutes a practical design principle for low-power implantable BCIs.

4.2. Limitations of the Ridge Decoder and Future Directions

The Ridge baseline ($r_{\text{neural}} = 0.831$) falls below the SNN ($r_{\text{neural}} = 0.932$) because the linear model cannot capture nonlinear temporal structures in motor cortex. Dynamic weighting (Eq. 6) partially compensates, but a performance gap remains. Future work should explore:

- LSTM or Transformer decoders for nonlinear temporal modelling.
- Systematic SER-robustness comparison across decoder complexities.
- Validation with the actual ASBIT channel model [1].
- Longer time-lag windows (> 200 ms) or spectral features.

4.3. Study Limitations

- **Project 1:** Synthetic data do not capture in-vivo noise structure or inter-channel correlations; scalability beyond 4 channels is unverified.
- **Project 2:** SER is modelled as i.i.d. Bernoulli, ignoring the burst-error patterns of real wireless channels; results are from a single subject, limiting cross-subject generalisation.

5. Conclusion

This paper quantifies two key claims for low-power BCI design:

1. An event-driven decoding pipeline maintains $R^2 \geq 0.91$ while reducing CPU active time by **91.35%** relative to 50 ms binning.
2. A Ridge Regression decoder achieves a performance retention rate of **0.9988** at $\text{SER} = 10^{-2}$, greatly exceeding the SNN benchmark (≈ 0.700).
3. The combination of event-driven acquisition and Ridge decoding simultaneously satisfies the accuracy, energy-efficiency, and error-robustness requirements of implantable wireless BCIs.

Acknowledgements

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